Project Records

Mithra D K

Overview

- **Aim**: Cluster songs into categories using MFCC data
- **Process:**
    1. Dataset and MFCC introduction
    2. Exploratory Data Analysis (EDA)
    3. Data Preprocessing
    4. Clustering and Evaluation
    5. Results and Recommendations

Introduction of MFCC files

1. **MFCC**: Represents short-term power spectrum of sound.
2. **Sampling rate**: 44100 Hz, hop size: 512 samples.
3. **File structure**:
    - 20 rows (MFCC coefficients).
    - Columns vary based on song duration (each column = time frame).
1. **Dataset**: 116 MFCC files from 6 song categories.

Our aim is to cluster these songs into the given six categories and identify each category as well.

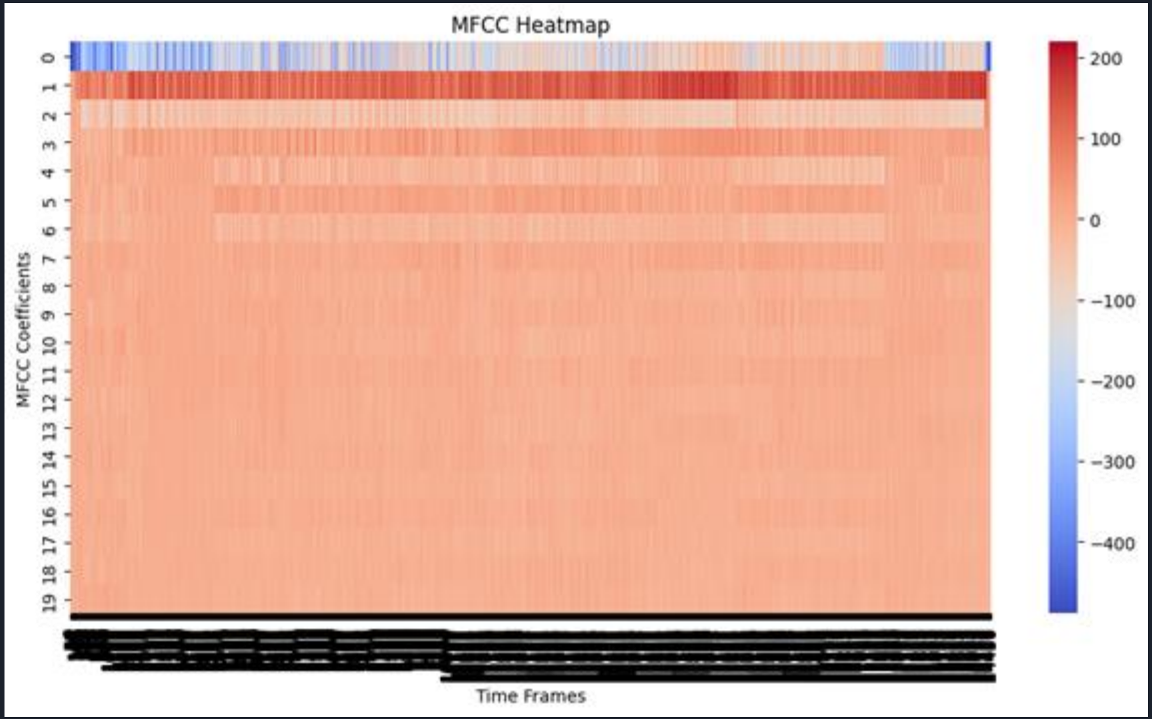
Exploratory Data Analysis (EDA)

Data Shape: It is observed that files vary in frames but have 20 coefficients each

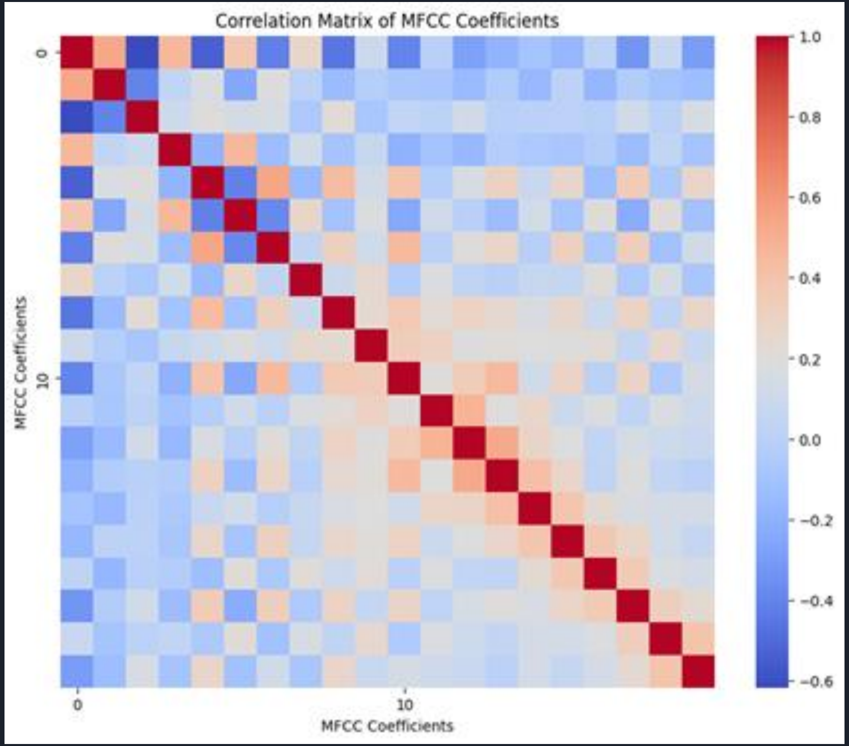
For visualization of MFCCs of individual songs, we used:

1. **Heat Map**
2. **Correlation Matrix**
3. **Histograms**

Heat Map: Heat map of a MFCC file helps us visualize the variation of each coefficient with time.

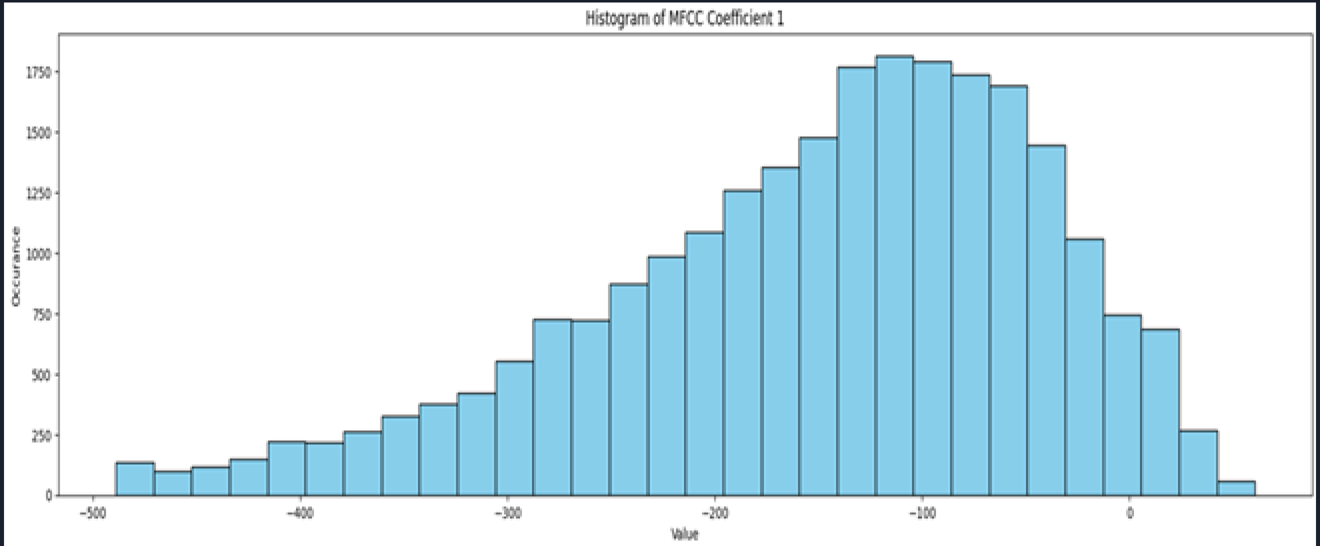


Correlation Matrix: For a specific MFCC, a correlation matrix helps us visualize correlation of coefficients. It shows that there is a chance of dimension reduction.

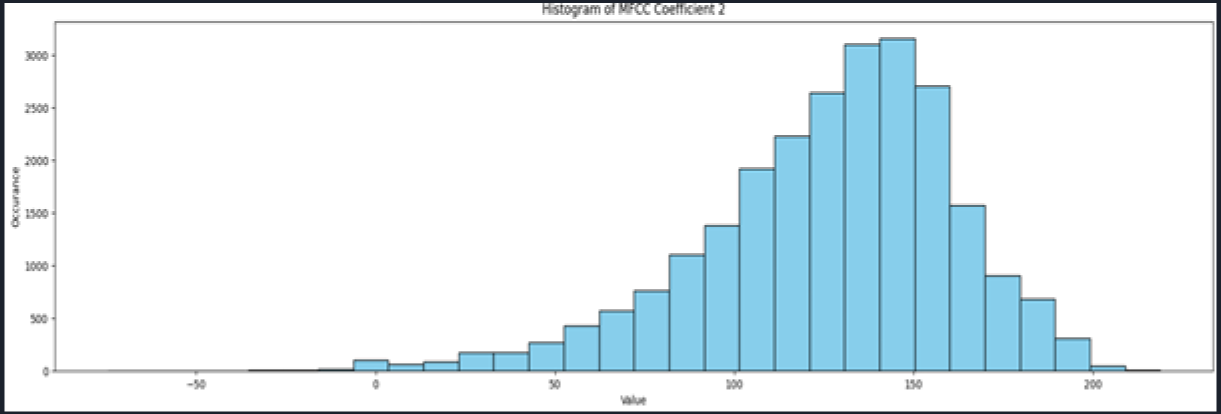


Histograms: For a specific MFCC, histograms of each coefficients help us see clearly the variance of each coefficient and its range. These histograms show us that there is a need of normalization too.

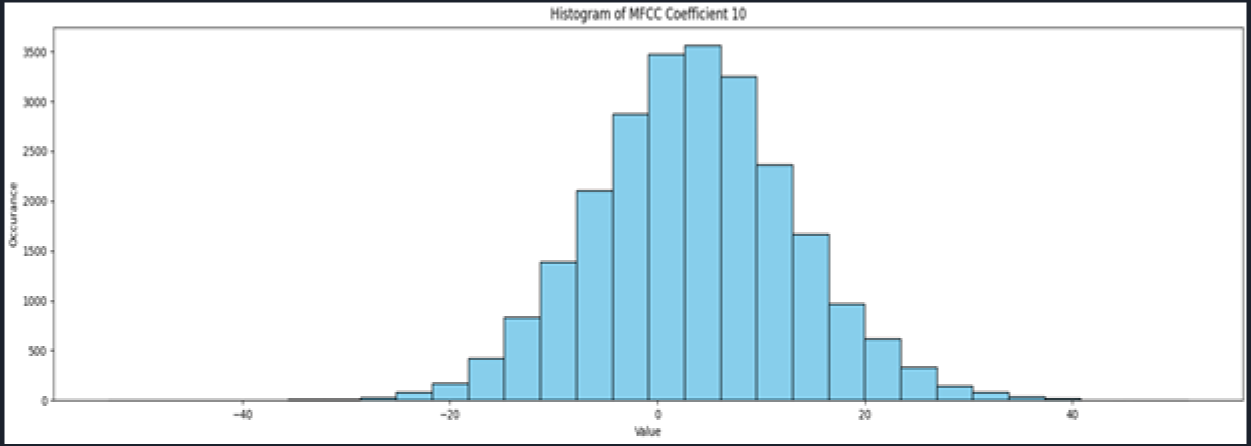
Coefficient 1:



Coefficient 2:



Coefficient 10:



Data Preprocessing

Dimensionality Reduction with PCA:

- UMAP is applied to reduce the dimensionality of each MFCC file.
- Captures the variation of each MFCC coefficient across the song.
- Reduces the data to a shape of **(20, 1)** for each song, preserving significant patterns.

Song Representation:

- After PCA, each song is represented as a **vector**.
- Collecting these vectors for all 116 songs forms a **data frame** of shape **(116, 20)**.
    - **Rows**: 116 songs.
    - **Columns**: 20 reduced dimensions (PCA features).

new_data

	0	1	2	3	4	5	6	7	
0	-24265.404107	15512.984143	850.999379	1988.268433	1359.292879	1470.881867	441.174611	1148.604515	
0	-23765.771299	13745.790477	3.748504	3540.851612	983.657379	2941.902285	-1459.600752	995.545805	
0	-27328.435468	18298.558312	-5406.741519	3909.388594	-156.500450	2366.565749	-836.749520	1658.954365	
0	-36040.257416	13011.451271	-5386.821445	7455.467886	-106.353432	4232.958828	-828.078596	3609.776458	
0	-36262.793151	30637.290731	-5657.867755	1787.230990	4924.062064	-378.983483	775.520206	536.284248	
...	...	...	...	...	...	...	...	...	
0	-17968.400285	11694.970880	-896.545230	2928.062271	524.351079	2108.790940	-95.249016	2027.118617	-
0	-42723.433391	35545.685106	-5899.016168	-1899.623659	6075.942555	-270.873251	-865.550638	1525.369673	
0	-37823.998456	35663.598814	-6721.440025	-1396.303073	4994.304265	457.942392	220.533748	1350.680159	
0	-26728.031691	23379.541243	-266.297086	11897.310639	-5322.646009	5005.971820	-394.376769	809.685699	-1
0	-27878.400482	16613.330949	-7009.020311	8326.133771	661.152687	1836.646738	2021.191013	526.516399	-1

116 rows × 20 columns

- **Need for Standardization**:
    - **Exploratory Data Analysis (EDA)** using histograms showed variations in scale across coefficients.
    - Data frame **(new_data)** is standardized to ensure better clustering.
    - Resulting standardized data frame: **new_data_1**.
- **Further Dimensionality Reduction with t-SNE**:
    - **Correlation Matrix** indicates possible redundancy in coefficients.
    - **t-SNE** is used to reduce dimensions further.
    - **Hyperparameters** like number of components and perplexity are tuned to optimize clustering results.

new_data_1

	0	1	2	3	4	5	6	7	8	9
0	1.147007	-0.991866	1.546269	-1.081888	-0.058625	-0.142366	0.182230	-0.307206	0.533198	-0.137476
1	1.216409	-1.272655	1.297190	-0.574224	-0.207283	0.848499	-1.492257	-0.428740	0.434126	-1.702746
2	0.721533	-0.549266	-0.293415	-0.453720	-0.658501	0.460958	-0.943556	0.098031	0.884647	0.572706
3	-0.488592	-1.389334	-0.287559	0.705777	-0.638656	1.718142	-0.935918	1.647058	0.067549	1.894229
4	-0.519504	1.411233	-0.367242	-1.147623	1.352134	-1.388417	0.476772	-0.793411	0.040280	-0.193731
...	...	...	...	...	...	...	...	...	...	...
111	2.021699	-1.598509	1.032517	-0.774594	-0.389054	0.287323	-0.290332	0.390368	-1.239132	0.443272
112	-1.416927	2.191128	-0.438137	-2.353151	1.807991	-1.315595	-0.968929	-0.008040	0.537185	-0.174776
113	-0.736365	2.209863	-0.679917	-2.188575	1.379932	-0.824672	-0.012143	-0.146750	0.078373	-0.630792
114	0.804933	0.258051	1.217800	2.158171	-2.703006	2.238835	-0.553848	-0.576320	-1.617912	-1.652020
115	0.645140	-0.817032	-0.764461	0.990467	-0.334915	0.104009	1.574145	-0.801167	-1.805321	1.062605

116 rows × 20 columns

Clustering Methods

- **Methods Tried**:
    - K-Means
    - Gaussian Mixture Models (GMM)
    - Hierarchical Clustering

Best performance: **Hierarchical (Agglomerative) Clustering**

Evaluation

K-Means clustering

- Silhouette Score: 0.2357251417977238 (Favourable: 1)

GMM

- Silhouette Score: 0.2357251417977238 (Favourable: 1)
- BIC Score: 30901.338142345616 (Favourable: Close to 0)

Hierarchical (Agglomerative) clustering

- Silhouette Score: 0.6150247 (Favourable: 1)
- Davies-Bouldin Index: 0.3878636180163831 (Favourable:Close to 0)

Conclusion

- Successfully clustered 116 songs into six categories using MFCC data.
- **Hierarchical (Agglomerative) Clustering** emerged as the best-performing method.
- **PCA** and **standardization** improved clustering quality by optimizing feature representation.
- Evaluation metrics confirmed Hierarchical Clustering effectiveness:
  - **Silhouette Score**: 0.615 (close to 1 is favorable)
  - **Davies-Bouldin Index**: 0.388 (closer to 0 is favorable)

Learnings

1. Files of very big sizes could be split and zipped into various zip files
2. The importance of exploratory data analysis for insights
3. Heat-map analysis
4. Dimension reduction techniques
5. Hyperparameter tuning
6. Appropriate selection of evaluation metric